

Supplementary material: A formal knowledge base for amplicon sequencing data and geochemistry of the Rub' al Khali desert

S1 Knowledge-representation patterns

The graph examples in this section instantiate the formal patterns in the main article with observations from the resource.

Prepend Table S1's declarations when parsing the Turtle snippets. The unprefix knowledge-base namespace is used by the PowerSoil protocol. The listing validator checks examples against the available module snapshot; full validation additionally requires the corresponding taxonomy payload and pinned reference terms.

Table S1 Prefix declarations used in all Turtle and SPARQL listings.

Prefix	Namespace
rak:	https://rubalkhali.science/kb/RAK_
rak_kb:	https://rubalkhali.science/kb/
sio:	http://semanticscience.org/resource/SIO_
owl:	http://www.w3.org/2002/07/owl#
pato:	http://purl.obolibrary.org/obo/PATO_
uo:	http://purl.obolibrary.org/obo/UO_
envo:	http://purl.obolibrary.org/obo/ENVO_
ncbitaxon:	http://purl.obolibrary.org/obo/NCBITaxon_
obo:	http://purl.obolibrary.org/obo/
rdf:	http://www.w3.org/1999/02/22-rdf-syntax-ns#
rdfs:	http://www.w3.org/2000/01/rdf-schema#
xsd:	http://www.w3.org/2001/XMLSchema#
geo:	http://www.opengis.net/ont/geosparql#
dcterms:	http://purl.org/dc/terms/

S1.1 Laboratory-control worked example

The control module represents the whole-cell D6300 product, its composition specification and evidence-bearing expected-taxon assertions as separate entities. The

specification links directly to each supported taxon; a reified assertion additionally preserves the subject, relation, taxon, status and source. Manufacturer genomic-DNA mass fractions and control occurrences retain their source provenance. The distributed control module links these materials to their processing roles and expected taxa (main article, Data records).

S1.2 Sampling pattern

A sampling event is a process that derives a sample from a feature of interest (the site).

Listing S1 is generated from the Trip 1, Site 1 deep-soil observation.

Listing S1 Turtle representation of a sampling process.

```
rak:P100001 a rak:0000024 ; # Sampling Process
  sio:000291 rak:1000001 ; # hasTarget
  sio:000229 rak:7000001 . # hasOutput (Collection)

rak:7000001 a sio:001419 ; # Material sample collection
  sio:000059 rak:6000001 . # hasMember

rak:6000001 a rak:0000021 ; # Deep Soil Sample
  sio:000244 rak:1000001 . # isDerivedFrom
```

The instance `rak:P100001` of a sampling process targets the site `rak:1000001` and yields a collection `rak:7000001` as its output. The collection explicitly contains `rak:6000001`, and this deep-soil sample is asserted to be derived from the same site. The redundancy between `hasTarget` on the process and `isDerivedFrom` on the sample is intentional: the former records the action, the latter records the resulting provenance edge directly on the material entity, so queries can traverse from sample to site without joining through the sampling event.

S1.3 Experimental protocol pattern

A protocol is a reusable information specification. A protocol execution is a process specified by that protocol. Each process (e.g., DNA extraction, library preparation) consumes inputs and produces outputs, explicitly linking agents and devices. The pattern instantiates SIO's general *Process-hasInput/hasOutput-Object* idiom and adds a specification link to a reusable protocol description.

Listing S2 demonstrates the formalization of a DNA extraction process.

Listing S2 Turtle representation of a DNA extraction protocol.

```
rak:P200001 a sio:000994 ; # Scientific Protocol Execution
  sio:000339 rak_kb:DNA_ext_powersoil ; # isSpecifiedBy
  sio:000230 rak:6000001 ; # hasInput (Soil Sample)
  sio:000229 rak:6800001 . # hasOutput (DNA Extract)

rak:6800001 a rak:0000040 ; # DNA Extract
  sio:000008 rak:5900001 . # hasAttribute (Concentration Quality)
```

The protocol-execution instance `rak:P200001` is specified by the named extraction protocol `rak_kb:DNA_ext_powersoil`, takes the soil sample `rak:6000001` as input, and produces the DNA extract `rak:6800001` as output; the resulting extract carries a concentration quality (`rak:5900001`, class `rak:0000043`) which is itself quantified through

a separate measurement process following the measurement pattern in the main article's Knowledge representation subsection. The measurement value that quantifies it is the separate individual `rak:4900001` (class `rak:0000044`), which points to the quality with `sio:000215` and back from it with `sio:000216`. Separate value and quality individuals preserve the reciprocal measurement links.

S1.4 Taxonomic abundance pattern

Taxonomic profiles contain read counts and relative read abundances aggregated by rank. The validated mapping assigns NCBI or project identifiers to each rank-truncated classifier lineage. Listing S3 gives one such observation.

For instance, the sequence analysis for a specific profile yields an absolute sequence count associated with an audited taxonomic class. We construct an abundance quality that is an attribute of both that taxon and the source sequence dataset, binding the count back to the biological workflow. The pattern uses SIO's *Collection-hasMember-Thing* and reciprocal *MeasurementValue-isMeasurementValueOf-Quality* idioms.

Listing S3 Turtle representation of taxonomic abundance formalized from a feature table.

```
rak:P290001 a rak:0000071 ; # Bioinformatic Workflow
  sio:000339 rak:L000012 ; # 16S amplicon processing protocol
  sio:000230 rak:7791009 ; # hasInput (FASTQ Dataset, ERR16783308)
  sio:000229 rak:7740001 . # hasOutput (Taxonomic Dataset)

rak:7740001 a rak:0000074 ; # taxon absolute abundance dataset
  sio:000059 rak:V00000001 . # hasMember (Observation Value)

rak:V00000001 a rak:0000076 ; # Absolute abundance value
  sio:000215 rak:Q00000001 ; # isMeasurementValueOf
  rak:2000025 "Domain: Bacteria" ; # classifier lineage
  rak:2000026 "853814.0"^^xsd:double . # absolute read count

rak:Q00000001 a rak:0000078, pato:0000070 ; # Absolute Abundance Quality
  sio:000011 ncbitaxon:2, rak:7791009 ; # taxon and FASTQ bearers
  sio:000216 rak:V00000001 . # hasMeasurementValue
```

The bioinformatic-workflow instance `rak:P290001` consumes the FASTQ dataset of run `ERR16783308` (`rak:7791009`, profile `V10Dr1`) and produces the absolute taxonomic dataset `rak:7740001`; that dataset has as one of its members the measurement value `rak:V00000001`, which records an absolute read count of 853,814 for the domain-rank record `Domain: Bacteria`. Its reciprocal quality `rak:Q00000001` is an attribute of both *Bacteria* (`ncbitaxon:2`) and the source FASTQ dataset. Each of the fourteen datasets produced by one workflow instance holds a single rank at one of the two abundance bases, so a rank-restricted query never has to filter observations by lineage depth. Each taxon-rank entry adds a value and a quality while referencing shared workflow, dataset and taxon objects. Contextual lineages use the same pattern with their audited project IRI.

S1.5 XRF geochemistry pattern

We model every XRF session as a process with multiple concentration-value outputs and a workflow-specific protocol. A laboratory session has an archived soil specimen

as input. The field instrument log identifies a test and site, and the graph links each session to that site. Field-XRF and laboratory-XRF records remain distinguishable even when they report the same analyte. Analyte cross-references to ChEBI¹ and PubChem² are maintained in a single mapping. The identifier audit resolves every non-null identifier against pinned ChEBI v247 and a dated PubChem snapshot, checking identifier existence, label, formula, charge and atom-versus-oxide entity type. Of 93 instrument channels, 79 have a supported ChEBI cross-reference and 91 have a formula-matching PubChem cross-reference. Missing external identifiers remain explicitly null. The `rdfs:seeAlso` annotations identify chemical references for reporting channels. The P_2O_5 channel reports oxide-equivalent magnitudes; its ChEBI cross-reference P_4O_{10} shares the empirical ratio. Light Elements is an instrument-defined aggregate channel with null chemical cross-references.

The shared output pattern specializes the measurement pattern (main article, Knowledge representation); only the laboratory specialization requires a soil-sample input.

Listing S4 is generated from the aluminium output of laboratory-XRF process `rak:P300072` in the XRF module.

Listing S4 Turtle representation of an XRF geochemistry measurement.

```
rak:P300072 a rak:0000025 ; # XRF Measuring Process
sio:000230 rak:6001732 ; # hasInput (sample V1Sr1)
sio:000339 rak:L_Surface_Lab ; # laboratory protocol
sio:000229 rak:4302121 . # hasOutput

rak:4302121 a rak:0000507 ; # Al concentration value
sio:000215 rak:5302121 ; # isMeasurementValueOf
rak:2000012 "0.78"^^xsd:double . # Concentration

rak:5302121 a rak:0000107 ; # Al concentration quality
sio:000011 rak:6001732 ; # isAttributeOf
sio:000216 rak:4302121 . # hasMeasurementValue
```

This laboratory-XRF process consumes archived specimen `rak:6001732` (sample V1Sr1) and produces an aluminium concentration value. The value points to its aluminium quality using `sio:000215`; the quality points back using `sio:000216`. A dedicated scan of every XRF source artifact (the processed laboratory table, the field log and all raw instrument exports) finds no unit declaration attached to any concentration column; the only unit token in the exports is PPM, and it belongs to the lower-limit-of-detection column. The generated module therefore carries no unit assertion for XRF concentrations. These values are instrument-reported magnitudes with undocumented concentration units. Field sessions use `sio:000291` to identify their site and retain the instrument test identifier; laboratory sessions identify their specimen input. Cross-workflow calibration requires documented units, paired aliquots and reference materials.

S1.6 Sequencing workflow pattern

Sequencing data are produced by a chain of laboratory and computational processes that the knowledge graph represents as a sequence of *Process-hasInput/hasOutput-Object* steps: a DNA extraction process consumes a soil sample and produces a DNA

extract; a library preparation process consumes the DNA extract and produces an amplicon library; a sequencing process consumes the amplicon library and produces a FASTQ dataset whose members are sequence reads. Each process is specified by a protocol and may carry agents and device participants, reusing the experimental protocol pattern in the main article.

Listing S5 instantiates the complete sequencing chain.

Listing S5 Turtle representation of the sequencing workflow from soil sample T1Dr1 to ENA run ERR16061083.

```
rak:P200923 a sio:000994 ; # DNA Extraction Process
  sio:000339 rak_kb:DNA_ext_powersoil ; # isSpecifiedBy
  sio:000230 rak:6000577 ; # hasInput (T1Dr1 Soil Sample)
  sio:000229 rak:6800540 . # hasOutput (DNA Extract)

rak:P250001 a rak:0000065 ; # Library Preparation Process
  sio:000339 rak:L000010 ; # Illumina 16S Prep Guide
  sio:000230 rak:6800540 ; # hasInput (DNA Extract)
  sio:000229 rak:6690001 . # hasOutput (Amplicon Library)

rak:P280001 a rak:0000066 ; # Sequencing Process
  sio:000230 rak:6690001 ; # hasInput (Amplicon Library)
  sio:000229 rak:7790001 . # hasOutput (FASTQ Dataset)

rak:7790001 a rak:0000063 ; # FASTQ Dataset
  rdfs:seeAlso <https://identifiers.org/insdc.run/ERR16061083> .
```

The DNA extraction process `rak:P200923` consumes the soil sample `rak:6000577` and produces the DNA extract `rak:6800540`; that extract is the input of the library preparation `rak:P250001`, which yields the amplicon library `rak:6690001`; finally the sequencing process `rak:P280001` consumes the library and produces the FASTQ dataset `rak:7790001`, which is cross-referenced to its ENA accession via `rdfs:seeAlso`. Each link in the chain is the same `hasInput/hasOutput` pattern, so a single SPARQL traversal can recover the recorded provenance from any node in this chain.

S1.7 Sequencing quality control pattern

Sequencing quality control is represented as an additional computational process that takes a FASTQ dataset as input and produces multiple measurement values, each instantiating the measurement pattern in the main article’s Knowledge representation subsection. Metrics (sequence count, GC content, duplicate rate, read length) are split into forward (R1) and reverse (R2) subclasses so that the provenance of paired-end runs is preserved.

Listing S6 shows the corresponding quality-control pattern.

Listing S6 Turtle representation of forward-read GC content for ENA run ERR16062319.

```
rak:P350001 a rak:0000200 ; # Sequencing QC Process
  sio:000230 rak:7790263 ; # hasInput (FASTQ Dataset)
  sio:000229 rak:4450001 . # hasOutput (QC Value)

rak:4450001 a rak:0000217 ; # Forward GC Content Value
  sio:000215 rak:5550001 ; # isMeasurementValueOf
  sio:000221 uo:0000187 ; # Percent
```

```
rak:2000074 "58.0"^^xsd:double . # has GC content
rak:5550001 a rak:0000205 ; # Forward GC Content Quality
sio:000011 rak:7790263 ; # isAttributeOf (FASTQ Dataset)
sio:000216 rak:4450001 . # hasMeasurementValue
```

The QC process `rak:P350001` takes the FASTQ dataset `rak:7790263` as input and produces the measurement value `rak:4450001`, which records a GC content of 58.0 percent and is the measurement value of a forward-read GC-content quality `rak:5550001` that is in turn an attribute of the same FASTQ dataset. Splitting the quality class into forward and reverse subclasses preserves the read direction, so that paired-end QC metrics can be queried independently per orientation.

S2 Library preparation and archived-soil pH

First-round PCR used 50 μL reactions with $1\times \text{Mg}^{2+}$ -containing buffer, 0.2 mM of each dNTP, 0.4 μM of each primer and 1.25 U of TaKaRa Ex Taq. The programme comprised 95°C for 3 min, followed by 35 cycles of 95°C for 10 s, 65°C for 30 s and 72°C for 1 min, with a final extension at 75°C for 5 min. Initial DNA input was 5 μL ; repeat attempts used 8 and 12 μL while retaining the 50 μL total reaction volume. Indexed libraries underwent 0.8:1 AMPure XP cleanup. The retained indexing protocol specifies the kit and thermal programme; its reaction-specific reagent volumes remain undocumented.

Pooling used library molarity derived from measured DNA concentration and fragment length. Aliquot volumes were calculated using a pooling-volume factor of the form VC/N , where V is target pool volume, C is target molarity and N is the number of libraries; dividing this factor by a library's molarity gives that library's aliquot volume. Run-specific target volumes, target molarities and individual library measurements remain undocumented.

Both PMA-treated and untreated aliquots underwent 10 min in darkness and 35 min of illumination at 450 nm. Each of three parent collection tubes supplied three separately prepared slurries, with each slurry split into a treated and an untreated aliquot.

Archived-soil pH suspensions were mixed for 30 min on a HulaMixer, then stood for 30 min at room temperature. Measurements used an accumet AB315 meter and Fisher Scientific accumet electrode 13-620-112, with the green stability indicator as the reading endpoint. All Trip 4 assays used archived field replicate 2. Each admitted measurement is linked to its physical specimen and retains its original assay identifier and value.

S3 XRF acquisition evidence

Laboratory XRF used a Vanta handheld analyser in Fast Mode for 547 specimens from Trips 1–4 and 178 from Trip 5. The Trip 5 acquisition settings were Fast Screening-He8mm, helium, an 8 mm diameter and oxide reporting. These specimens comprise 60 shallow-subsurface, 60 surface and 58 root-adjacent samples. For V4Dr3 and V18Dr3,

the analytical dataset uses Fast Screening measurements; additional Best Detection measurements are retained as acquisition data.

The exact method label, gas and diameter for Trips 1–4 remain undocumented. Handheld field measurements used Geochem(3-Beam) and NORMAL settings on an Olympus Vanta analyser; acquisition duration was unrecorded.

The concentration observations retain their instrument-reported magnitudes. Percentages associated with statistical error and units attached to detection limits describe those quantities. Concentration units, the exact laboratory model variant and serial number, exposure settings, sample preparation and calibration results remain undocumented. Acquisition validation checks specimen selection, repeated observations and the unit evidence described in the main article.

S4 Supplementary tables

Table S2 reports the XRF channel cross-references. The complete field environmental observations are distributed as tabular data (main article, Data records).

Table S2: Audited XRF reporting-channel cross-references. Labels, formulas, charge and entity types were checked against pinned ChEBI and PubChem records.

Analyte	Formula/type	ChEBI	PubChem
Ag	Ag	CHEBI:30512	CID:23954
Al	Al	CHEBI:28984	CID:5359268
Al2O3	Al2O3	CHEBI:30187	CID:9989226
As	As	CHEBI:27563	CID:5359596
Au	Au	CHEBI:29287	CID:23985
Ba	Ba	CHEBI:32594	CID:5355457
BaO	BaO	–	CID:62392
Bi	Bi	CHEBI:33301	CID:5359367
Br	Br	CHEBI:22927	CID:5360770
Ca	Ca	CHEBI:22984	CID:5460341
CaO	CaO	CHEBI:31344	CID:14778
Cd	Cd	CHEBI:22977	CID:23973
Ce	Ce	CHEBI:33369	CID:23974
CeO2	CeO2	CHEBI:79089	CID:73963
Cl	Cl	CHEBI:23116	CID:5360523
Co	Co	CHEBI:27638	CID:104730
CoO	CoO	–	CID:14786
Cr	Cr	CHEBI:28073	CID:23976
Cr2O3	Cr2O3	CHEBI:48242	CID:517277
Cs	Cs	CHEBI:30514	CID:5354618
Cu	Cu	CHEBI:28694	CID:23978
CuO	CuO	CHEBI:75955	CID:164827
Dy	Dy	CHEBI:33377	CID:23912
Er	Er	CHEBI:33379	CID:23980
Eu	Eu	CHEBI:32999	CID:23981
Fe	Fe	CHEBI:18248	CID:23925

Analyte	Formula/type	ChEBI	PubChem
Fe2O3	Fe2O3	CHEBI:50819	CID:518696
Ga	Ga	CHEBI:49631	CID:5360835
Ga2O3	Ga2O3	–	CID:158605
Gd	Gd	CHEBI:33375	CID:23982
Ge	Ge	CHEBI:30441	CID:6326954
Hf	Hf	CHEBI:33343	CID:23986
Hg	Hg	CHEBI:25195	CID:23931
Ho	Ho	CHEBI:49648	CID:23988
I	I	CHEBI:24859	CID:5360629
Ir	Ir	CHEBI:49666	CID:23924
K	K	CHEBI:26216	CID:5462222
K2O	K2O	CHEBI:88321	CID:9989219
LE	instrument_pseudoanalyte	–	–
La	La	CHEBI:33336	CID:23926
La2O3	La2O3	–	CID:150906
Lu	Lu	CHEBI:33382	CID:23929
Mg	Mg	CHEBI:25107	CID:5462224
MgO	MgO	CHEBI:31794	CID:14792
Mn	Mn	CHEBI:18291	CID:23930
MnO	MnO	–	CID:14940
Mo	Mo	CHEBI:28685	CID:23932
Na	Na	CHEBI:26708	CID:5360545
Na2O	Na2O	–	CID:73971
Nb	Nb	CHEBI:33344	CID:23936
Nd	Nd	CHEBI:33372	CID:23934
Nd2O3	Nd2O3	–	CID:159373
Ni	Ni	CHEBI:28112	CID:935
NiO	NiO	–	CID:14805

Analyte	Formula/type	ChEBI	PubChem
Os	Os	CHEBI:30687	CID:23937
P	P	CHEBI:28659	CID:5462309
P2O5	P2O5	CHEBI:37376	CID:6326975
Pb	Pb	CHEBI:25016	CID:5352425
PbO	PbO	CHEBI:81045	CID:14827
Pr	Pr	CHEBI:49828	CID:23942
PrO2	PrO2	—	—
Pt	Pt	CHEBI:33364	CID:23939
Rb	Rb	CHEBI:33322	CID:5357696
Re	Re	CHEBI:49882	CID:23947
S	S	CHEBI:26833	CID:5362487
SO3	SO3	CHEBI:29384	CID:24682
Sb	Sb	CHEBI:30513	CID:5354495
Sc	Sc	CHEBI:33330	CID:23952
Sc2O3	Sc2O3	—	CID:134661
Se	Se	CHEBI:27568	CID:6326970
Si	Si	CHEBI:27573	CID:5461123
SiO2	SiO2	CHEBI:30563	CID:24261
Sm	Sm	CHEBI:33374	CID:23951
Sm2O3	Sm2O3	—	CID:159425
Sn	Sn	CHEBI:27007	CID:5352426
Sr	Sr	CHEBI:33324	CID:5359327
SrO	SrO	—	CID:73975
Ta	Ta	CHEBI:33348	CID:23956
Tb	Tb	CHEBI:33376	CID:23958
Th	Th	CHEBI:33385	CID:23960
Ti	Ti	CHEBI:33341	CID:23963
TiO2	TiO2	CHEBI:32234	CID:26042

Analyte	Formula/type	ChEBI	PubChem
Tm	Tm	CHEBI:33380	CID:23961
U	U	CHEBI:27214	CID:23989
V	V	CHEBI:27698	CID:23990
V2O5	V2O5	CHEBI:30045	CID:14814
W	W	CHEBI:27998	CID:23964
Y	Y	CHEBI:33331	CID:23993
Yb	Yb	CHEBI:33381	CID:23992
Zn	Zn	CHEBI:27363	CID:23994
ZnO	ZnO	CHEBI:36560	CID:3007857
Zr	Zr	CHEBI:33342	CID:23995
ZrO2	ZrO2	—	CID:62395

Interpretation. A dash denotes an audited null. These `rdfs:seeAlso` cross-references identify the element or oxide-equivalent reporting channel. The P_2O_5 channel is cross-referenced to the ChEBI P_4O_{10} class only because the formulas share the P:O empirical ratio. LE is an instrument aggregate, and PrO_2 has no formula-matching external identifier in the audited sources.

References

- [1] Degtyarenko, K., Matos, P., Ennis, M., Hastings, J., Zbinden, M., McNaught, A., Alcántara, R., Darsow, M., Guedj, M., Ashburner, M.: ChEBI: a database and ontology for chemical entities of biological interest. *Nucleic Acids Research* **36**(suppl_1), 344–350 (2008) <https://doi.org/10.1093/nar/gkm791>
- [2] Kim, S., Chen, J., Cheng, T., Gindulyte, A., He, J., He, S., Li, Q., Shoemaker, B.A., Thiessen, P.A., Yu, B., *et al.*: PubChem 2023 update. *Nucleic Acids Research* **51**(D1), 1373–1380 (2023)